

University of Dhaka
Department of Computer Science & Engineering
CSE 3216: Software Design Pattern Lab (Batch 28)

Assignment 2 - Pattern-based Refactoring

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1 Identify and Justify Design Patterns (5 marks)

1.1 Builder Pattern

1.1.1 Purpose and UML Diagram

The **Builder Pattern** is a creational design pattern that separates the construction of a complex object from its representation, allowing the same construction process to create different representations.

UML Diagram:

1.1.2 Problem Solved

Location: SBackend/services/containerBuilder.js

The Builder pattern solves the problem of constructing complex Docker container configurations with numerous optional parameters. Without this pattern, we would face:

1. **Telescoping Constructor Anti-pattern:** A constructor with 20+ parameters would be unmaintainable.
2. **Inflexibility:** Different container types would require separate constructors.
3. **Readability Issues:** Passing multiple configuration objects would be confusing.

Example Before (Without Builder):

Listing 1: Problematic Approach

```
const config = {
  Image: 'project-sandbox:latest',
  name: 'sandbox-${projectId}',
  Hostname: 'sandbox',
  AttachStdin: true,
  AttachStdout: true,
  // ... 50+ more configuration options
  HostConfig: {
    Memory: 1024 * 1024 * 1024,
    MemorySwap: 1024 * 1024 * 1024,
    CpuShares: 1024,
    // ... more nested config
  },
  // ... deeply nested structure
};
```

Example After (With Builder):

Listing 2: Clean and Flexible Approach Using Builder Pattern

```
const builder = new SandboxContainerBuilder();
const director = new ContainerDirector(builder);
const containerConfig = director.buildSandboxContainer(
  projectId,
  containerName,
  workspacePath,
```

```
'project-sandbox:latest',
{
  containerMemory: 1024 * 1024 * 1024,
  containerCpu: 1.0
}
);
```

Benefits:

- **Method Chaining:** Fluent interface improves readability.
- **Step-by-step Construction:** Complex objects are built incrementally.
- **Reusability:** Common patterns encapsulated in methods like `createSandboxConfiguration()`.
- **Validation:** Validation logic can be added before invoking `build()`.
- **Modularity:** Each builder method handles a specific configuration aspect, improving maintainability.
- **Extensibility:** New builders (e.g., `DevelopmentContainerBuilder`, `ProductionContainerBuilder`) can be added without modifying existing code, adhering to the Open/Closed Principle.

1.2 State Pattern

1.2.1 Purpose and UML Diagram

The State pattern is a behavioral design pattern that allows an object to alter its behavior when its internal state changes, making the object appear to change its class.

1.2.2 Problem Solved

Location: `SBackend/services/containerStates.js`

The State pattern solves the problem of managing container lifecycle states (stopped, running, paused, error, removed) without using complex conditional logic.

Example Before (Without State Pattern):

Listing 3: Problematic Approach with Conditional Logic

```
class ContainerManager {
  async start(container) {
    if (this.state === 'stopped') {
      await container.start();
      this.state = 'running';
    } else if (this.state === 'paused') {
      await container.unpause();
      this.state = 'running';
    } else if (this.state === 'running') {
      throw new Error('Already running');
    } else if (this.state === 'error') {
      // Try to recover...
      // Complex recovery logic mixed with state management
    }
  }
}
```

```

    }
    // ... more conditions
}

async stop(container) {
    if (this.state === 'running') {
        // ... stop logic
    } else if (this.state === 'paused') {
        // ... unpause then stop logic
    }
    // ... more conditions
}
// Each method becomes a complex switch statement
}

```

Example After (With State Pattern):

Listing 4: Clean and Maintainable State Pattern Implementation

```

class ContainerWrapper {
    async start() {
        return this.state.start(); // Delegates to current state
    }

    async stop() {
        return this.state.stop(); // Each state knows its behavior
    }
}

class RunningState extends ContainerState {
    async stop() {
        await this.containerWrapper.dockerContainer.stop({ t: 10 });
        this.containerWrapper.setState(new StoppedState(this.containerWrapper));
    }
}

```

Benefits:

- **Eliminates Conditionals:** No more if-else chains for state transitions.
- **Single Responsibility:** Each state class handles one state's behavior.
- **Open/Closed Principle:** Add new states without modifying existing code.
- **Type Safety:** Each state can have different method signatures.
- **Event Emission:** State transitions emit events for monitoring.

1.3 Repository Pattern

1.3.1 Purpose and UML Diagram

The Repository pattern acts as a mediator between business logic and data access, providing an abstraction over data storage that allows the application to interact with data

through a collection-like interface.

1.3.2 Problem Solved

Current Implementation: Backend/crud.py and Backend/routers/projects.py

Note: The Repository pattern has been implemented in Backend/repositories/project_repository with both SQL and in-memory implementations. However, the routers currently still use CRUD classes directly. The refactoring shown below demonstrates how the routers should be updated to use the repository pattern.

The Repository pattern will:

1. **Decouple Data Access:** Business logic won't depend on SQLAlchemy directly.
2. **Improve Testability:** Can inject mock repositories for unit testing.
3. **Centralize Data Logic:** All database operations in one place.
4. **Enable Multiple Data Sources:** Can switch between database, cache, or file storage.

Example Before (Current Implementation):

Listing 5: Direct CRUD Usage - Tight Coupling

```
# Backend/routers/projects.py
@router.post("/", response_model=schemas.Project)
async def create_project(
    project_in: schemas.ProjectCreate,
    db: AsyncSession = Depends(get_db)
):
    # Direct CRUD usage - tight coupling
    owner = await crud.crud_user.get(db, id=project_in.owner_id)
    if not owner:
        raise HTTPException(status_code=404, detail="Owner not found")

    return await crud.crud_project.create(db, obj_in=project_in)
```

Example After (With Repository Pattern):

Listing 6: Refactored Implementation Using Repository Pattern

```
# Backend/repositories/project_repository.py
from abc import ABC, abstractmethod
from typing import Optional
from uuid import UUID
from sqlalchemy.ext.asyncio import AsyncSession
from sqlalchemy import select
from models import Project
import schema as schemas

class IProjectRepository(ABC):
    @abstractmethod
```

```

    async def get_by_id(self, project_id: UUID) ->
Optional[Project]:
    pass

    @abstractmethod
    async def create(self, project: schemas.ProjectCreate) ->
Project:
    pass

class SQLProjectRepository(IPProjectRepository):
    def __init__(self, db: AsyncSession):
        self.db = db

    async def get_by_id(self, project_id: UUID) ->
Optional[Project]:
        result = await self.db.execute(
            select(Project).where(Project.project_id == project_id)
        )
        return result.scalar_one_or_none()

    async def create(self, project: schemas.ProjectCreate) ->
Project:
        project_data = project.model_dump()
        db_project = Project(**project_data)
        self.db.add(db_project)
        await self.db.commit()
        await self.db.refresh(db_project)
        return db_project

# Backend/routers/projects.py (Refactored)
from fastapi import APIRouter, Depends, HTTPException, status
from uuid import UUID
import schema as schemas
from repositories import IPProjectRepository
from repositories.dependencies import get_project_repository

@router.post("/", response_model=schemas.Project,
    status_code=status.HTTP_201_CREATED)
async def create_project(
    project_in: schemas.ProjectCreate,
    repo: IPProjectRepository = Depends(get_project_repository)
):
    # Business logic separated from data access
    # Note: Owner validation would need a UserRepository as well
    return await repo.create(project_in)

```

Benefits:

- **Testability:** Easy to mock repositories in unit tests.
 - **Flexibility:** Can swap implementations (SQL, NoSQL, in-memory).
-

- **Maintainability:** Centralized data access logic.
- **Domain-Driven Design:** Aligns with DDD principles.

2 Refactor and Implement (7 marks)

2.1 Repository Pattern Refactoring

Before Architecture

(Architecture diagram or description placeholder)

Implementation

Step 1: Create Repository Interface and Implementation

The repository interface and implementations already exist in Backend/repositories/project_reposi
The structure is shown below.

Listing 7: Repository Interface and SQL Implementation

```
# Backend/repositories/project_repository.py
from abc import ABC, abstractmethod
from typing import Optional, List
from uuid import UUID
from sqlalchemy.ext.asyncio import AsyncSession
from sqlalchemy import select, func
from models import Project
import schema as schemas

class IProjectRepository(ABC):
    @abstractmethod
    async def get_by_id(self, project_id: UUID) ->
Optional[Project]:
        pass

    @abstractmethod
    async def get_all(self, skip: int = 0, limit: int = 100,
**filters) -> List[Project]:
        pass

    @abstractmethod
    async def create(self, project: schemas.ProjectCreate) ->
Project:
        pass

    @abstractmethod
    async def update(self, project_id: UUID, project:
schemas.ProjectUpdate) -> Project:
        pass

    @abstractmethod
    async def delete(self, project_id: UUID) -> bool:
```



```

        pass

    @abstractmethod
    async def count(self, **filters) -> int:
        pass

class SQLProjectRepository(IPProjectRepository):
    def __init__(self, db: AsyncSession):
        self.db = db

    async def get_by_id(self, project_id: UUID) ->
Optional[Project]:
        result = await self.db.execute(
            select(Project).where(Project.project_id == project_id)
        )
        return result.scalar_one_or_none()

    async def get_all(self, skip: int = 0, limit: int = 100,
**filters) -> List[Project]:
        query = select(Project)
        for key, value in filters.items():
            if hasattr(Project, key) and value is not None:
                query = query.where(getattr(Project, key) == value)
        query = query.offset(skip).limit(limit)
        result = await self.db.execute(query)
        return result.scalars().all()

    async def create(self, project: schemas.ProjectCreate) ->
Project:
        project_data = project.model_dump()
        db_project = Project(**project_data)
        self.db.add(db_project)
        await self.db.commit()
        try:
            await self.db.refresh(db_project)
        except Exception:
            db_project = await
self.get_by_id(db_project.project_id)
        return db_project

    async def update(self, project_id: UUID, project:
schemas.ProjectUpdate) -> Project:
        db_project = await self.get_by_id(project_id)
        if not db_project:
            raise ValueError(f"Project {project_id} not found")
        update_data = project.model_dump(exclude_unset=True)
        for field, value in update_data.items():
            setattr(db_project, field, value)
        self.db.add(db_project)
        await self.db.commit()
        try:

```

```

        await self.db.refresh(db_project)
    except Exception:
        db_project = await self.get_by_id(project_id)
    return db_project

async def delete(self, project_id: UUID) -> bool:
    db_project = await self.get_by_id(project_id)
    if not db_project:
        return False
    await self.db.delete(db_project)
    await self.db.commit()
    return True

```

Step 2: Create Dependency Injection

The dependency injection is defined in `Backend/repositories/dependencies.py`.

Listing 8: Dependency Injection for Repository

```

# Backend/repositories/dependencies.py
from fastapi import Depends
from sqlalchemy.ext.asyncio import AsyncSession
from db import get_db
from repositories.project_repository import SQLProjectRepository,
    IProjectRepository

async def get_project_repository(
    db: AsyncSession = Depends(get_db)
) -> IProjectRepository:
    return SQLProjectRepository(db)

```

Step 3: Refactor Router (Planned Implementation)

The routers currently use CRUD classes directly. The planned refactoring is shown below.

Listing 9: Planned Router Refactoring with Repository Pattern

```

# Backend/routers/projects.py
from fastapi import APIRouter, Depends, HTTPException, status
from uuid import UUID
import schema as schemas
from repositories import IProjectRepository
from repositories.dependencies import get_project_repository

router = APIRouter(prefix="/projects", tags=["projects"])

@router.post("/", response_model=schemas.Project,
    status_code=status.HTTP_201_CREATED)
async def create_project(
    project_in: schemas.ProjectCreate,
    repo: IProjectRepository = Depends(get_project_repository)
):

```

```

        return await repo.create(project_in)

@router.get("/{project_id}", response_model=schemas.Project)
async def read_project(
    project_id: UUID,
    repo: IProjectRepository = Depends(get_project_repository)
):
    project = await repo.get_by_id(project_id)
    if not project:
        raise HTTPException(
            status_code=status.HTTP_404_NOT_FOUND,
            detail="Project not found"
        )
    return project

```

Implementation Status:

- Repository interface and implementations created (Backend/repositories/project_repository)
- Dependency injection function created (Backend/repositories/dependencies.py)
- Router refactoring planned but not yet implemented

Benefits When Fully Implemented:

- Separation of concerns
- Testability (mock repositories can be injected)
- Flexibility (swap implementations easily)
- Maintainability (centralized data access)

2.2 Builder Pattern Refactoring

Before Architecture (Without Builder Pattern)

Location: SBackend/services/containerService.js

Problems:

- Massive configuration object (100+ lines of nested configuration)
- No reusability or modularity
- Difficult to test configuration
- Tight coupling between service and configuration logic
- No validation layer

Example Before (Without Builder Pattern):

Listing 10: Massive Inline Configuration Before Refactoring

```

async createContainer(projectId) {
    const container = await this.docker.createContainer({
        Image: this.IMAGE_NAME,

```

```

name: containerName,
Hostname: 'sandbox',
AttachStdin: true,
AttachStdout: true,
AttachStderr: true,
Tty: true,
OpenStdin: true,
StdinOnce: false,
Env: [
  'TERM=xterm-256color',
  'NODE_ENV=development',
  'PROJECT_ID=${projectId}',
  'SHELL=/bin/bash',
  'HOME=/home/developer',
  'USER=developer',
  'LANG=C.UTF-8',
  'LC_ALL=C.UTF-8'
],
WorkingDir: '/workspace',
User: 'root',
HostConfig: {
  ReadonlyRootfs: false,
  CapDrop: ['ALL'],
  CapAdd: ['CHOWN', 'SETUID', 'SETGID', 'DAC_OVERRIDE'],
  SecurityOpt: ['no-new-privileges'],
  Memory: this.config.containerMemory,
  MemorySwap: this.config.containerMemory,
  CpuShares: Math.floor(this.config.containerCpu * 1024),
  PidsLimit: 512,
  NetworkMode: 'bridge',
  Binds: ['${workspacePath}:/workspace:rw'],
  PortBindings: {
    '22/tcp': [{ HostPort: '0' }],
    '3000/tcp': [{ HostPort: '0' }]
  },
  Tmpfs: { '/tmp': 'rw,size=100m' },
  Ulimits: [{ Name: 'nofile', Soft: 1024, Hard: 2048 }],
  AutoRemove: false
},
ExposedPorts: { '22/tcp': {}, '3000/tcp': {} },
Labels: {
  'project.id': projectId,
  'service': 'sandbox'
}
});
await container.start();
}

```

Example After (With Builder Pattern and State Pattern):

Listing 11: Builder Pattern and State Pattern Refactoring

```
async createContainer(projectId) {
  const builder = new SandboxContainerBuilder();
  const director = new ContainerDirector(builder);
  const containerConfig = director.buildSandboxContainer(
    projectId,
    containerName,
    workspacePath,
    this.IMAGE_NAME,
    {
      containerMemory: this.config.containerMemory,
      containerCpu: this.config.containerCpu
    }
  );

  const container = await this.docker.createContainer(
    containerConfig);

  const containerWrapper = new ContainerWrapper(container,
    projectId);
  await containerWrapper.initialize(workspacePath, null);
  await containerWrapper.start();
}
```

Benefits Achieved:

- Clear separation between construction and use of objects
- Reusable and modular configuration building
- Independent testing of configuration logic
- Centralized maintainability
- Extensible through new concrete builders (Open/Closed Principle)
- Integrated State Pattern for lifecycle management

2.3 State Pattern Refactoring

Before Architecture (Without State Pattern)

Location: SBackend/services/containerService.js (before refactoring)

Problems:

- Complex conditional logic with if/else chains for state transitions
- No state tracking — current container state unknown
- Error-prone — invalid operations may be called (e.g., start already running container)
- No validation — cannot check if operation is allowed before attempting
- Tight coupling — service directly manages all state transitions

Listing 12: Container Service before refactoring (without State Pattern)

```
// SBackend/services/containerService.js (OLD CODE)
async createContainer(projectId) {
  // ... container creation ...
  const container = await this.docker.createContainer(
    containerConfig);
  try {
    await container.start();
    console.log('Container started');
  } catch (error) {
    console.error('Failed to start:', error);
  }
}

async stopContainer(projectId) {
  const containerInfo = this.containers.get(projectId);
  if (!containerInfo) return;

  const container = this.docker.getContainer(containerInfo.id);
  try {
    const info = await container.inspect();
    if (info.State.Status === 'running') {
      await container.stop({ t: 10 });
      console.log('Container stopped');
    } else if (info.State.Status === 'exited') {
      console.log('Container already stopped');
    } else if (info.State.Status === 'paused') {
      await container.unpause();
      await container.stop({ t: 10 });
      console.log('Container unpaused and stopped');
    } else {
      console.log('Container in unknown state:', info.State.Status);
    }
  } catch (error) {
    console.error('Error stopping container:', error);
  }
}

async restartContainer(projectId) {
  // More if/else chains for restart logic
}
```

After Architecture (With State Pattern)

Example After (With State Pattern):

Listing 13: Container Service after applying State Pattern

```
// SBackend/services/containerService.js (REFACTORED CODE)
async createContainer(projectId) {
  const container = await this.docker.createContainer(
    containerConfig);
```

```

    const containerWrapper = new ContainerWrapper(container,
        projectId);
    await containerWrapper.initialize(workspacePath, null);
    await containerWrapper.start(); // Delegates to StoppedState.
        start()
}

async stopContainer(projectId) {
    const containerWrapper = this.containers.get(projectId);
    if (!containerWrapper) return;
    await containerWrapper.stop(); // Delegates to current state
}

async restartContainer(projectId) {
    const containerWrapper = this.containers.get(projectId);
    if (!containerWrapper) return;
    await containerWrapper.restart(); // Delegates to current state
}

```

Benefits Achieved:

- Eliminates conditional logic for state transitions
- Maintains accurate state tracking
- Prevents invalid operations through enforced state rules
- Allows pre-validation with `canStart()`, `canStop()`
- Separates concerns across state classes
- Emits events during state transitions for monitoring
- Provides automatic recovery via `ErrorState`

Listing 14: Chain of Responsibility usage example

```

result = permission_chain.has_permission(
    permission="canEdit",
    role_name="editor",
    role_permissions={"canEdit": True}
)

```

3 Reflection and Testing (3 marks)

3.1 Reflection: Improvements and Trade-offs

Improvements Achieved

1. **Modularity** – Data access isolated via repository interfaces
2. **Maintainability** – Patterns encapsulate complex logic
3. **Testability** – Mock repositories enable fast unit testing

4. **Scalability** – Open/Closed principle supported
5. **Readability** – Builder and State patterns simplify structure

Trade-offs

- Additional abstraction increases learning curve
- Slight performance overhead (negligible)
- More classes and interfaces required initially

Design Decisions

1. Repository Pattern prioritized for **Projects**
2. State Pattern applied to container lifecycle
3. Chain of Responsibility for permissions hierarchy
4. Builder Pattern for Docker container configuration

3.2 Testing Summary

Test Strategy

- Unit tests for isolated components
- Integration tests for pattern interactions
- Mock tests without real dependencies

Repository Pattern Tests

Listing 15: Repository pattern test cases

```
@pytest.mark.asyncio
async def test_repository_get_by_id(db_session):
    repo = SQLProjectRepository(db_session)
    project = await repo.create(schemas.ProjectCreate(
        project_name="Test Project",
        owner_id=uuid4(),
        visibility="private"
    ))
    found = await repo.get_by_id(project.project_id)
    assert found.project_name == "Test Project"

@pytest.mark.asyncio
async def test_repository_mock_implementation():
    repo = InMemoryProjectRepository()
    project = await repo.create(schemas.ProjectCreate(
        project_name="Mock Project",
        owner_id=uuid4(),
        visibility="private"
    ))
    assert project.project_name == "Mock Project"
```


State Pattern Tests

Listing 16: State pattern test cases

```
describe('Container State Pattern', () => {
  let mockContainer, wrapper;

  beforeEach(() => {
    mockContainer = { start: jest.fn(), stop: jest.fn(), inspect:
      jest.fn() };
    wrapper = new ContainerWrapper(mockContainer, 'test-project');
  });

  test('transition from stopped to running', async () => {
    await wrapper.setState(new StoppedState(wrapper));
    await wrapper.start();
    expect(mockContainer.start).toHaveBeenCalled();
  });

  test('should not start from running state', async () => {
    await wrapper.setState(new RunningState(wrapper));
    await expect(wrapper.start()).rejects.toThrow();
  });
});
```

Builder Pattern Tests

Listing 17: Builder pattern test cases

```
describe('Container Builder Pattern', () => {
  test('should build configuration with chaining', () => {
    const builder = new SandboxContainerBuilder();
    const config = builder
      .setImage('test-image')
      .setName('test-container')
      .setMemoryLimits(1024 * 1024 * 1024)
      .addPortBinding(3000)
      .build();
    expect(config.Image).toBe('test-image');
  });
});
```

Testing Status:

- Builder, State, and CoR fully implemented and tested
- Repository implemented, pending router integration
- Example test cases provided for all patterns

Functional Stability Verification

- Container lifecycle fully functional with State and Builder
- Permission checks validated with CoR

- Project CRUD stable under Repository abstraction

4 Conclusion

The application of four key design patterns—Builder, State, Repository, and Chain of Responsibility—significantly enhanced modularity, testability, and maintainability. Despite minor trade-offs in abstraction complexity, the system now exhibits robust scalability and architectural clarity.

Appendix: File Locations

- Builder Pattern: `SBackend/services/containerBuilder.js`
 - State Pattern: `SBackend/services/containerStates.js`
 - Repository Pattern: `Backend/repositories/project_repository.py`
-